Influence of Monetary and Fiscal Policy on Aggregate Demand

Introduction

- How do fiscal and monetary policy affect real variables in the long run?
- In short run, policymakers use FP and MP to stabilize fluctuations in output, unemployment and inflation.
- AD can shift due to policy action or changes in spending patterns by households and firms.
- *Note to students: Ignore some topics in this chapter which have so far not been covered in class (**).*

How Monetary Policy affects Aggregate Demand

- Interest effect is presumably strongest of all effects which explain by AD is downward sloping.
- At least for US economy.
- Need for a theory of interest rates
- Theory of Liquidity Preference
- Interest adjusts to balance supply and demand for money.

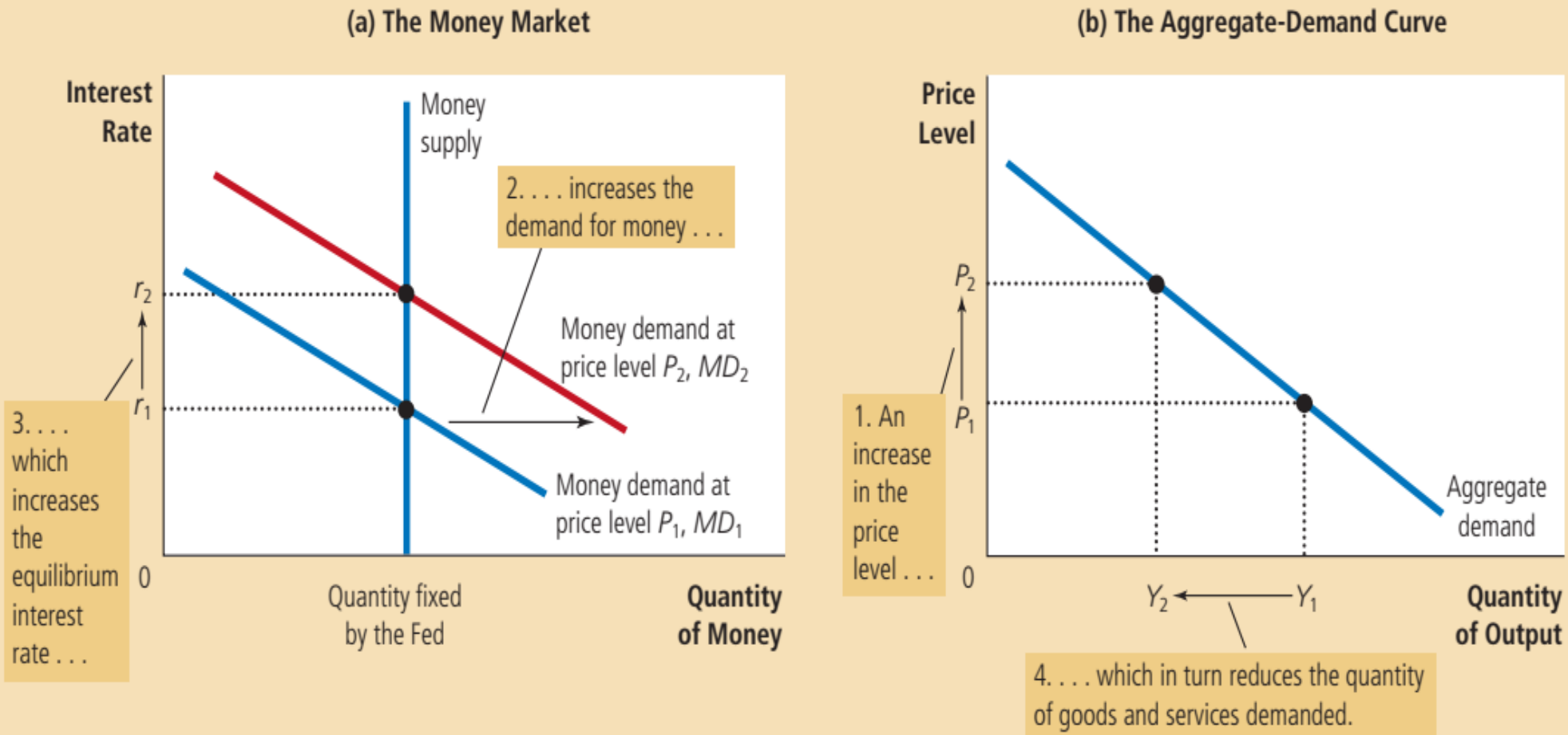
Theory of Liquidity Preference

- Nominal and Real IR are moving together; expected inflation and money supply are held constant.
- Money Demand is downward sloping, because interest rate is the opportunity cost of holding money.
- At higher prices, money demand is high, Md curve shifts upwards.

Figure 2

The Money Market and the Slope of the Aggregate-Demand Curve

An increase in the price level from P_1 to P_2 shifts the money-demand curve to the right, as in panel (a). This increase in money demand causes the interest rate to rise from r_1 to r_2. Because the interest rate is the cost of borrowing, the increase in the interest rate reduces the quantity of goods and services demanded from Y_1 to Y_2. This negative relationship between the price level and quantity demanded is represented with a downward-sloping aggregate-demand curve, as in panel (b).

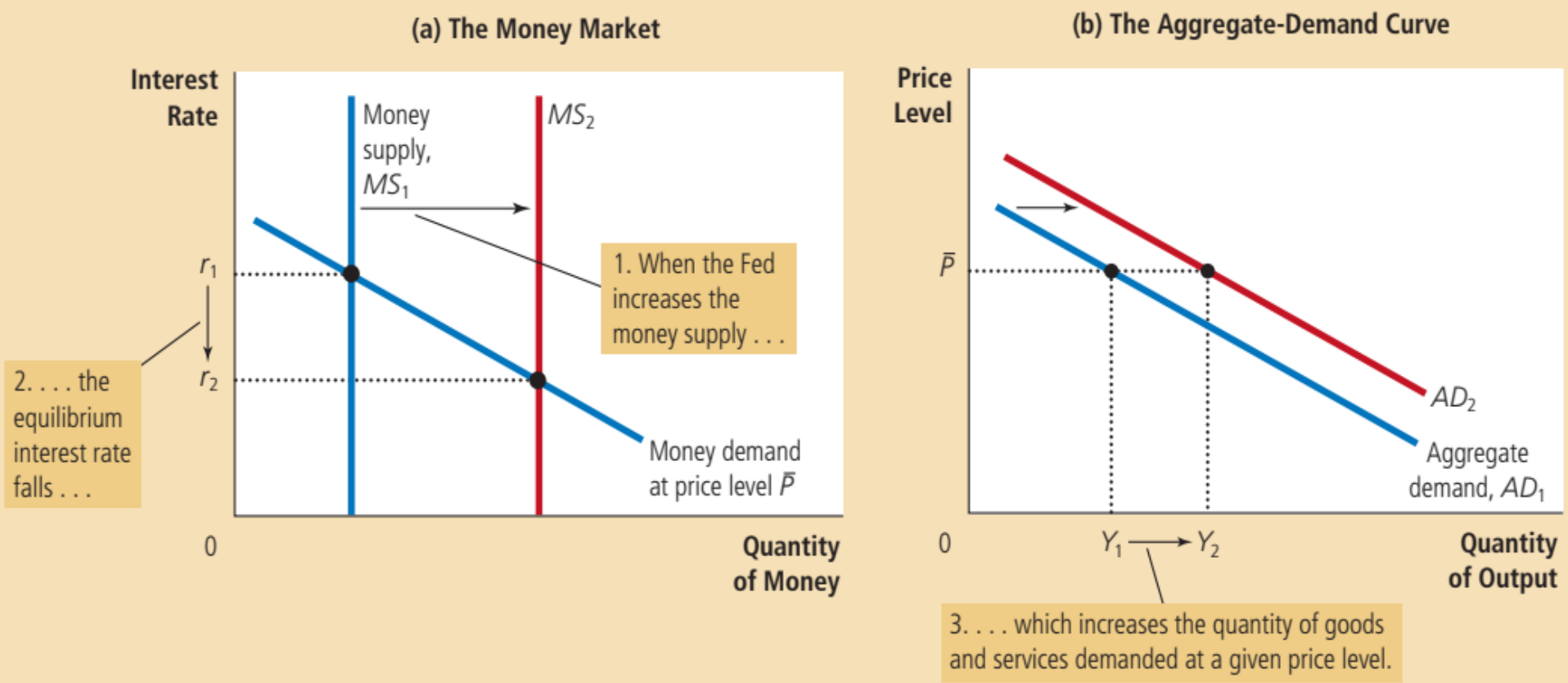


Changes in Money Supply

Figure 3

A Monetary Injection

In panel (a), an increase in the money supply from MS_1 to MS_2 reduces the equilibrium interest rate from r_1 to r_2. Because the interest rate is the cost of borrowing, the fall in the interest rate raises the quantity of goods and services demanded at a given price level from Y_1 to Y_2. Thus, in panel (b), the aggregate-demand curve shifts to the right from AD_1 to AD_2.



The Role of Interest Rate targets in Fed Policy (**)

- Fed usually uses interest rates rather than Money supply to moderate aggregate demand.
- Ms is hard to measure, and Md is fluctuating.
- Hence Fed uses IR, usually federal funds rate, to achieve target of money supply, via buying and selling of government bonds.
- MP can be described either in terms of money or interest rate.

How Fiscal Policy affects Aggregate Demand

- Changes in G shift AD curve.
- **Multiplier effect:** Initial spending increase has a cascading effect, so a G amount increase in government spending increases AD by more than G, and vice versa.
- Logic of ME also applies to changes from C or I or NX.
- Investment accelerator: positive feedback from AD to investment.

Multiplier effect

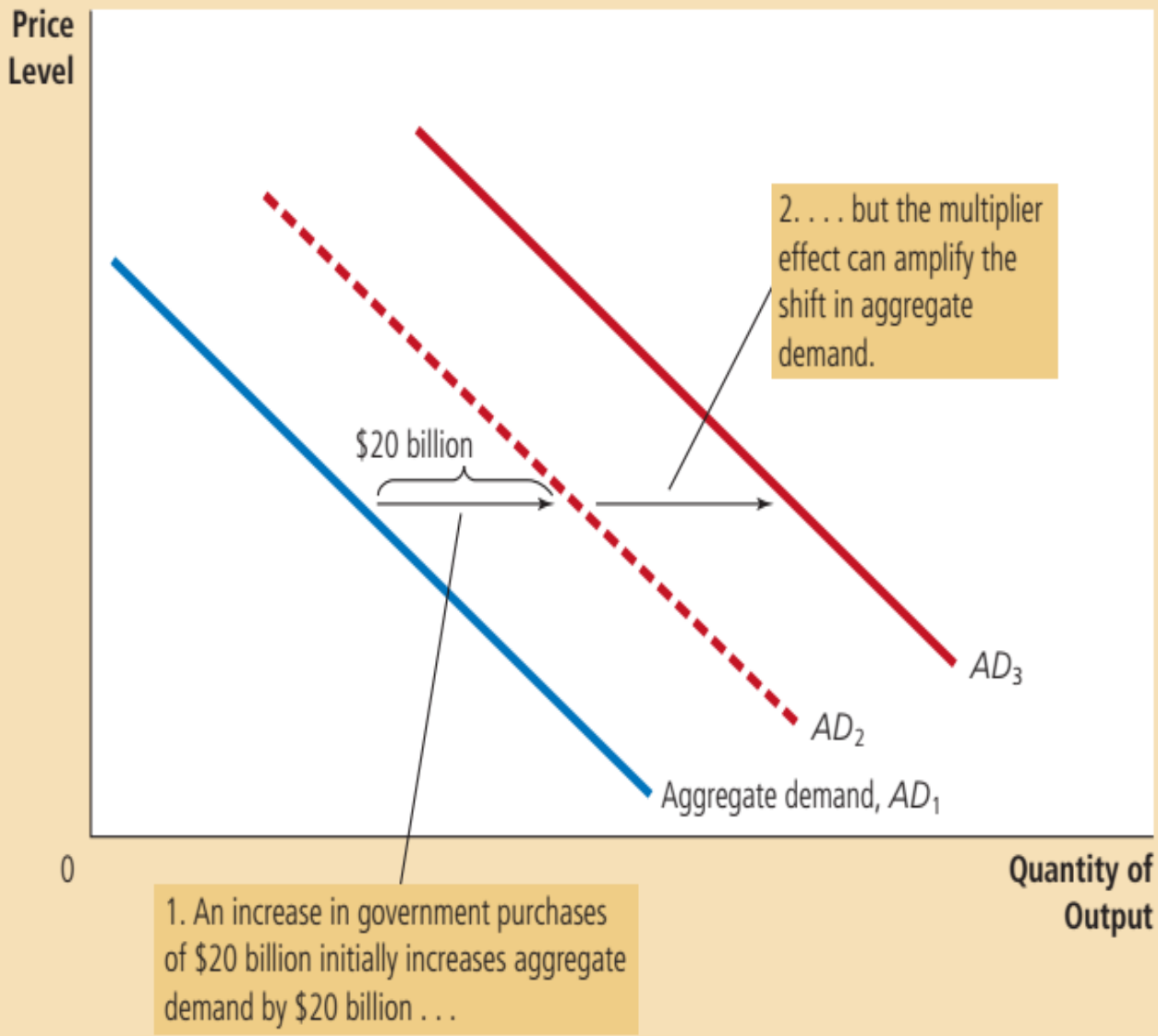


Figure 4

The Multiplier Effect

An increase in government purchases of $20 billion can shift the aggregate-demand curve to the right by more than $20 billion. This multiplier effect arises because increases in aggregate income stimulate additional spending by consumers.

Formula for spending multiplier

- MPC: marginal propensity to consume

To find the total impact on the demand for goods and services, we add up all these effects:

Change in government purchases	=	\$20 billion
First change in consumption	= MPC ×	\$20 billion
Second change in consumption	= MPC^2 ×	\$20 billion
Third change in consumption	= MPC^3 ×	\$20 billion
•		•
•		•
•		•

Total change in demand

$$= (1 + MPC + MPC^2 + MPC^3 + \ldots) \times \$20 \text{ billion.}$$

Multipler effect Formula

To simplify this equation for the multiplier, recall from math class that this expression is an infinite geometric series. For x between –1 and +1,

$$1 + x + x^2 + x^3 + \ldots = 1/(1 - x).$$

In our case, $x = MPC$. Thus,

$$\text{Multiplier} = 1/(1 - MPC).$$

For example, if MPC is ¾, the multiplier is $1/(1 - ¾)$, which is 4. In this case, the $20 billion of government spending generates $80 billion of demand for goods and services.

Crowding Out effect

- Counters multiplier effect, if G increases and ME causes AD to shift outward, it increases P, which raises Md. This raises IR and hurts private investment.

To sum up: *When the government increases its purchases by $20 billion, the aggregate demand for goods and services could rise by more or less than $20 billion depending on the sizes of the multiplier and crowding-out effects*. The multiplier effect by itself makes the shift in aggregate demand greater than $20 billion. The crowding-out effect pushes the aggregate-demand curve in the opposite direction and, if large enough, could result in an aggregate-demand shift of less than $20 billion.

Changes in Taxes (**)

- Changing taxes is another tool of government fiscal policy.
- Shift in AD from tax change is similarly affected by multiplier effect or crowding out effect.
- Temporary or permanent tax cut : If households *expect* tax cut to be permanent, spending might increase more. If it is *expected* to be temporary, households *expect* future increase in taxes, hence spending increase might not be too much.

Using Policy to stabilize the economy (**)

- **Case for Active Stabilization Policy:** Govts use fiscal and monetary policy to prevent fluctuations in economic variables, via moderating aggregate demand.
- Keynes: AD plays key role on short term fluctuations, bolstered by waves of irrational optimism or pessimism 'animal spirits'.
- 'Self fulfilling prophecy'

Using Policy to stabilize the economy (**)

- **Case against Active Stabilization Policy:** Govt policy making is not perfect, and certainly not instantaneous.
- Govt collects data after certain intervals, few days to few months to an year. Impossible to react quickly.
- Govt policies also react with a long lag. Multiplier or crowding effect takes months to factor in.
- Steady growth in Ms preferred.

Using Policy to stabilize the economy (**)

- **Automatic Stabilizers:** Automatic stabilizers are changes in fiscal policy that stimulate aggregate demand when the economy goes into a recession without policymakers having to take any deliberate action.
- Govt transfers like unemployment insurance increases during recessions, thus govt spending rises automatically.
- Argument on taxes seems sketchy!
- Balanced budget is not preferred.